

[Sankranti Joshi](#)

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Nationality: British | OCI Holder (India)

Professional Summary

- Machine Learning Engineer with 7+ years of experience in deep learning, computer vision, NLP, and MLOps.
- Worked on traditional ML, Deep Learning, Large Model training and inference, Visual prompt enhancing techniques, across a range of domains.

Technical Skills

- **Programming & Tools:** Python, PyTorch, TensorFlow, Bash, Docker, PyTorch-Lightning, Weights & Biases, Slurm, HPC, Poetry, scikit-learn, XGBoost, Hugging Face, Kubernetes (basic)
- **Machine Learning & AI:** Transformers, CNNs, RNN/LSTMs, MAE, VAE, Language Models, Contrastive Learning, Diffusion Models
- **Cloud & MLOps:** AWS (SageMaker, EC2, RDS), Azure, CI/CD, Model Versioning, Databricks

Industry & Research Experience

- **Senior Machine Learning Engineer | (July 2025 – Present) | Playstation**
 - Developed databricks pipelines to scale reinforcement learning and deep learning based training and inference pipelines
 - Developed models for HudReader (OCR detection across different languages)
 - Worked on defining prompt contexts and refining visual prompts using Set-of-Marks for Screen Deduplication in Anomaly detection
- **Senior Machine Learning Engineer | (March 2025 – May 2025) | Self-employed contract**
 - Worked for lean health startup (Digehealth) as the sole ML engineer
 - Built ML training and inference pipelines from scratch under limited infrastructure and noisy data conditions.
 - Boosted YOLO-based audio-segmentation performance by 15+ mAP points through data processing, noise cancellation and hyperparameter tuning.
 - Implemented reproducible training workflows on Azure ML with scalable, serverless compute.
 - Acted as the sole ML and infra owner in a founder-led startup, while engaging directly with Microsoft engineers to resolve GPU driver issues.
 - Factor analysis between YOLO predictions and downstream gut health related tasks.

- **PhD Researcher (Earth Observation & ML) | The University of Edinburgh, UK (Oct 2023 – Mar 2025)**
 - Developed multimodal contextual dataset combining Landsat and Sentinel data, migrated to [Amazon Open Data Sponsorship](#).
 - Developed distributed training pipelines and docker files to train on HPC and Kubernetes clusters.
 - Designed self-supervised MAE-based regression models for domain generalization in Earth Observation.
 - Investigated large-scale foundation models for geospatial vision tasks.
 - Withdrew due to realization that the time and situation wasn't right for a PhD.
- **Applied Science Intern (Internship as a requirement of the CDT) | Roku, Cambridge, UK (Jul 2024 – Sep 2024)**
 - Generated synthetic datasets using Stable Diffusion, ControlNet, inpainting/outpainting to enhance YOLOX model training.
 - Developed RAG pipelines that generate prompts for generating and fine-tuning prompts for diffusion models.
 - Led test/train dataset distribution strategy, improving model generalization.
- **Senior AIML Platform Engineer | GSK, London, UK (May 2023 - Oct 2023)**
 - Designed scalable ML pipelines for bio-model deployment on Azure HPC & cloud services.
 - Developed an end-to-end DNA sequence classification pipeline, showcasing ML platform capabilities, and investigated fine-tuning, pruning and compression techniques.
 - Transitioned to PhD program after internal role shift toward platform engineering.
- **Senior Machine Learning Engineer | Informa Markets, London, UK (Sep 2022 – Mar 2023)**
 - Developed LDA-guided BERT (LLM models) for document embeddings, improving NLP model accuracy by 25%.
 - Built MLOps infrastructure (data versioning, model versioning, dependency management, CI/CD).
 - Engineered dynamic pricing forecasting models using advanced data pipelines.
- **Senior Associate I, Data Science | AIG, London, UK (June 2019 – Aug 2022)**
 - Promoted from Machine Learning Engineer to Senior Associate I, Data Science (Apr - 2021).
 - Developed and maintained production repository for training and deployment of time-series forecasting model.
 - Built geospatial Django pipelines, migrating legacy systems to DVC-based versioning.

- Developed end-to-end feature-engineering pipelines in Python with MLflow integration, enabling fully reproducible data transformations and experiment tracking.
- Refactored existing MXNet implementation of DeepAR-based time-series forecasting for tailored use-case.
- Left following department divestiture; Declined transition to acquiring company.
- **Graduate Software Developer (NLP Research) | Sage Group Plc, Newcastle, UK (Feb 2018 – Mar 2019)**
 - Developed a linguist-verified ground-truth generation pipeline for user chat data, ensuring high-quality annotations.
 - Engineered and benchmarked custom NLP models (unsupervised methods, memory-augmented LSTMs, CRFs) against RASA and Alterra, driving measurable improvements in FAQ bot accuracy.
 - Moved on to pursue production-focused ML work and growth opportunities.

Key Projects & Contributions

- **Open Data Contribution:** Created an open-source, multi-sensor satellite dataset, now part of AWS Open Data Registry.
- **Deep Learning for Vision:** Implemented MAE-based contrastive and self-supervised learning models for domain adaptation in geospatial imagery.
- **Synthetic Data Generation:** Applied Diffusion models to generate high-quality synthetic datasets for AI training.
- **MLOps with CI/CD and reproducibility:** Built automated ML pipelines with CI/CD and reproduction.

Education

- **Master of Science in AI,** The University of Edinburgh, UK (2016 - 2017)
- **Master of Technology in Software Engineering,** Manipal Institute of Technology, India (2012 - 2014)
- **Bachelor of Technology in Computer Engineering,** Mody Institute of Technology and Science, India (2008 - 2012)